MID-PROJECT STATUS PRESENTATION

CarClinch Dealership Assistant

03/16/2026 · Group 4

1. PROJECT STATUS OVERVIEW

OVERALL STATUS

On Track (MVP)

Changes:

- **Blocker**: refactored email to basic **SMTP/IMAP**
- **LLM integration** & **Azure** deployment
- **Tightened scope** to remove certain nice-to-have's (phone assistant)

SCHEDULE

Sprint 2 -> Sprint 3

Up next:

- **PII sanitization**
- **Sentiment/intent analysis** on messages
- **Escalation** logic
- **Followup** timer configuration

Future sprints:

- Calendar integration
- Reporting/analytics

KEY MILESTONE

LLM Integration & Azure Deployment

By Wednesday:

- Demo **GPT 4.1 mini response** generation to client
- Demo various scenarios to test **guardrails on LLM behavior**
- Demo **Terraform** deployment to Azure 'dev' environment

2. COMPLETED TASKS & MILESTONES

COMPLETED TASKS

- Develop basic web form frontend (HTML/CSS/JS) and Azure Functions (Python) form backend API
- Refactored Azure Durable Functions (Python) email processing service to use SMTP/IMAP
- Establish & iterate on NoSQL database schema
- Deploy infrastructure to Azure dev environment with Terraform
- Integrate GPT-4.1 mini into Azure Durable Function when generating responses

ACTIVITY TIMELINE

Week 5 — **Design** core email system

Week 6 — Develop **form frontend/backend and email processing service**

Week 7-8 — Refactor email handling from ACS -> **SMTP/IMAP**

Week 9 — Integrate **GPT-4.1 mini** to generate responses & **Terraform** deployment to **Azure**

Week 10 — **Sentiment/intent** analysis, **escalation logic**, **followup rules**

3. CHALLENGES FACED & RESOLUTIONS

CHALLENGE	RESOLUTION	IMPACT
Blocker: ACS does not natively handle inbound emails	Refactored email handling to basic SMTP/IMAP polling on the inbox	Medium
Decentralized documentation led to team confusion	Centralized all documentation on a Google Docs, and all tasks onto GitHub issues	Low
Lack of dataset to train ML models for sentiment/intent analysis	Decided with client to use GPT-4.1 mini for both analytics and response generation	Medium

4. RESOURCE UTILIZATION

FINANCIAL RESOURCES

Budget Used: $0

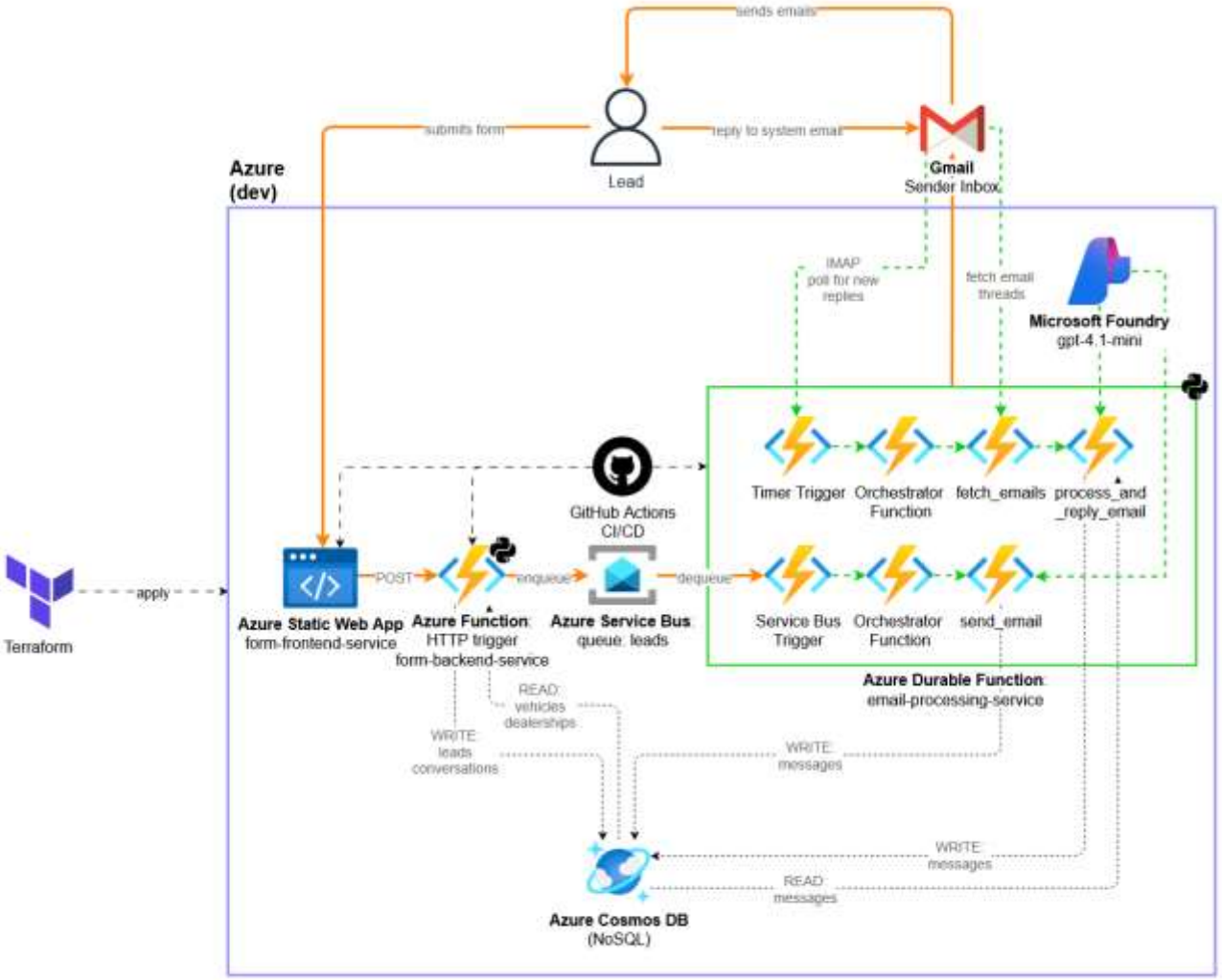
Local development when possible
- Docker Compose:
  - Docker Hub
  - Azure emulators

Free tiers or **student credits** used on Azure & GitHub

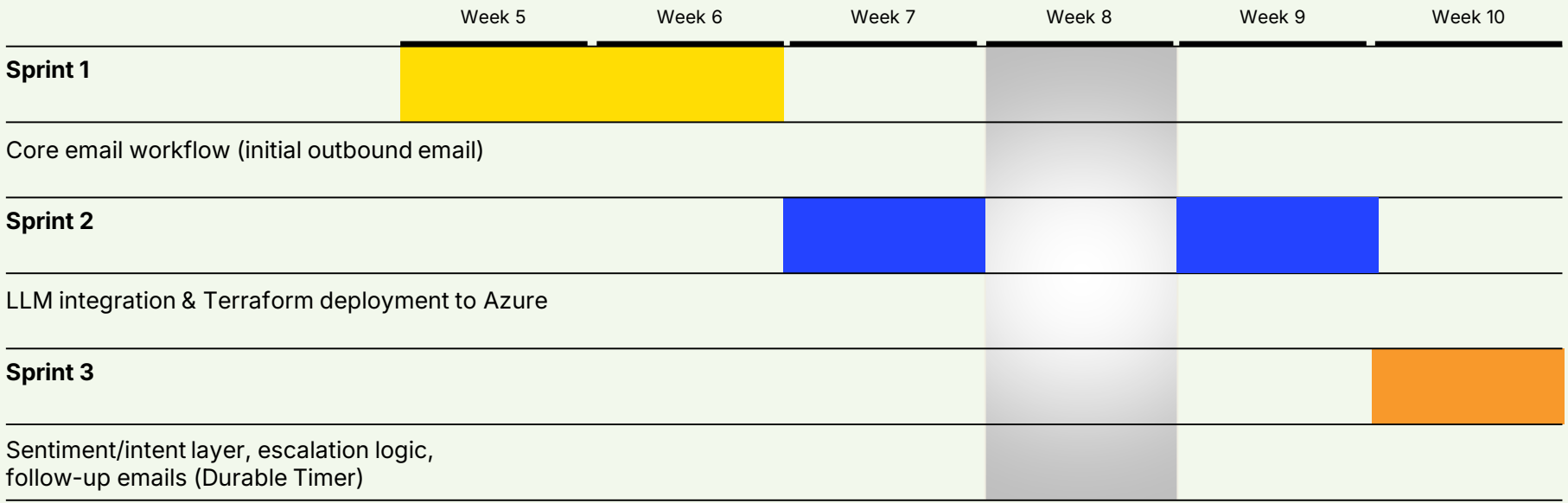
TECHNOLOGY & TOOLS

Tools in Use

Azure: Static Web App, App Service, Functions, Storage Account, Cosmos DB, Service Bus, Monitor, App Insights, Key Vault, Managed Identities (RBAC)
GitHub: Projects, Actions, Organizations
Docker: Docker Hub
Terraform: azurerm, github


sends emails
submits form
reply to system email
Lead
Gmail
Sender Inbox
Azure
(dev)
IMAP
poll for new
replies
fetch email
threads
Microsoft Foundry
gpt-4.1-mini
Timer Trigger
Orchestrator
Function
fetch_emails
process_and
_reply_email
GitHub Actions
CI/CD
Terraform
apply
POST
enqueue
dequeue
Azure Static Web App
form-frontend-service
Azure Function:
HTTP trigger
form-backend-service
Azure Service Bus
queue: leads
Service Bus
Trigger
Orchestrator
Function
send_email
Azure Durable Function:
email-processing-service
READ:
vehicles
dealerships
WRITE:
leads
conversations
WRITE:
messages
WRITE:
messages
READ:
messages
Azure Cosmos DB
(NoSQL)

Sprint Phases



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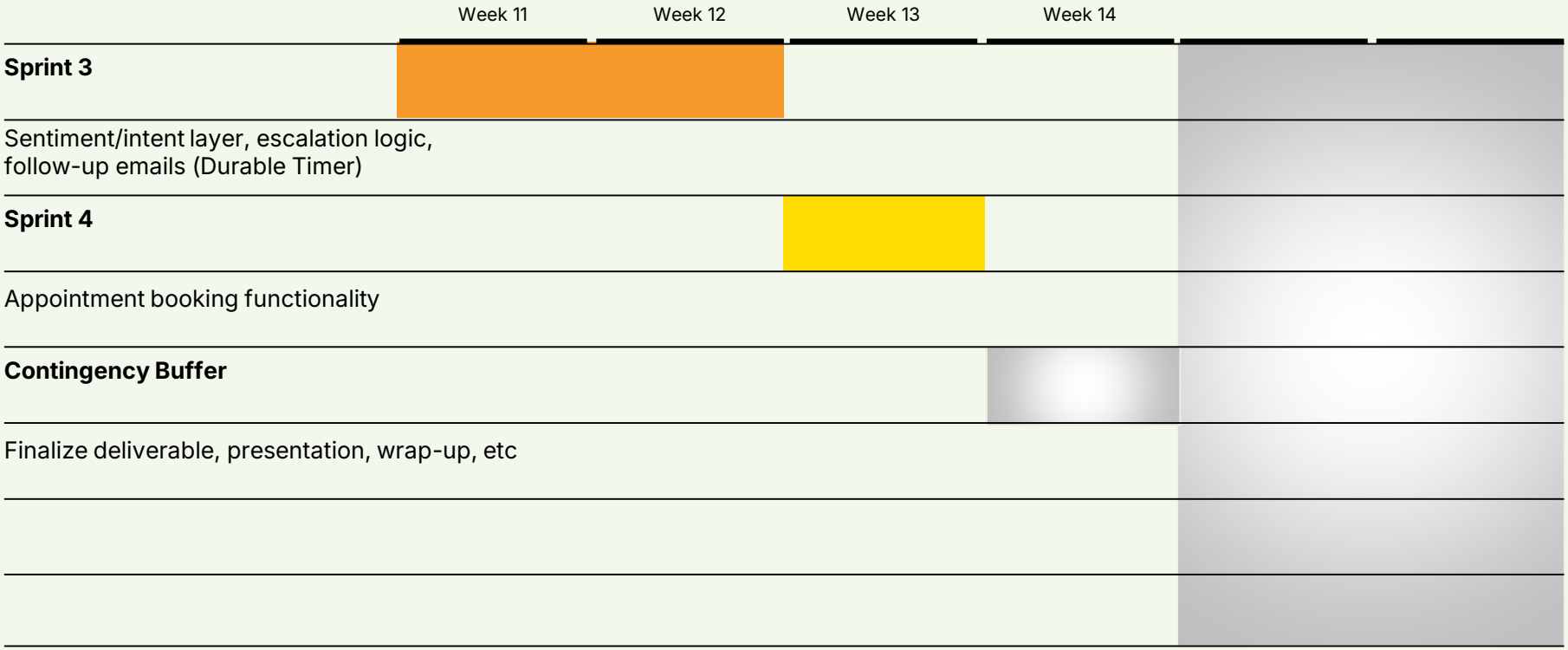

Live Demo!

5. NEXT STEPS & Q&A

UPCOMING PRIORITIES

- Alice: plan Sprint 3, distribute responsibilities per team member and feature development
- Aryan: secure Terraform/Bicep IaC configuration for staging/prod environment
- Desmond: finish testing sprint 2 features (QA) and feature development
- Junfeng: design and designate core feature development for Sprint 3
- Saizal: documentation support and feature development

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Any Questions?